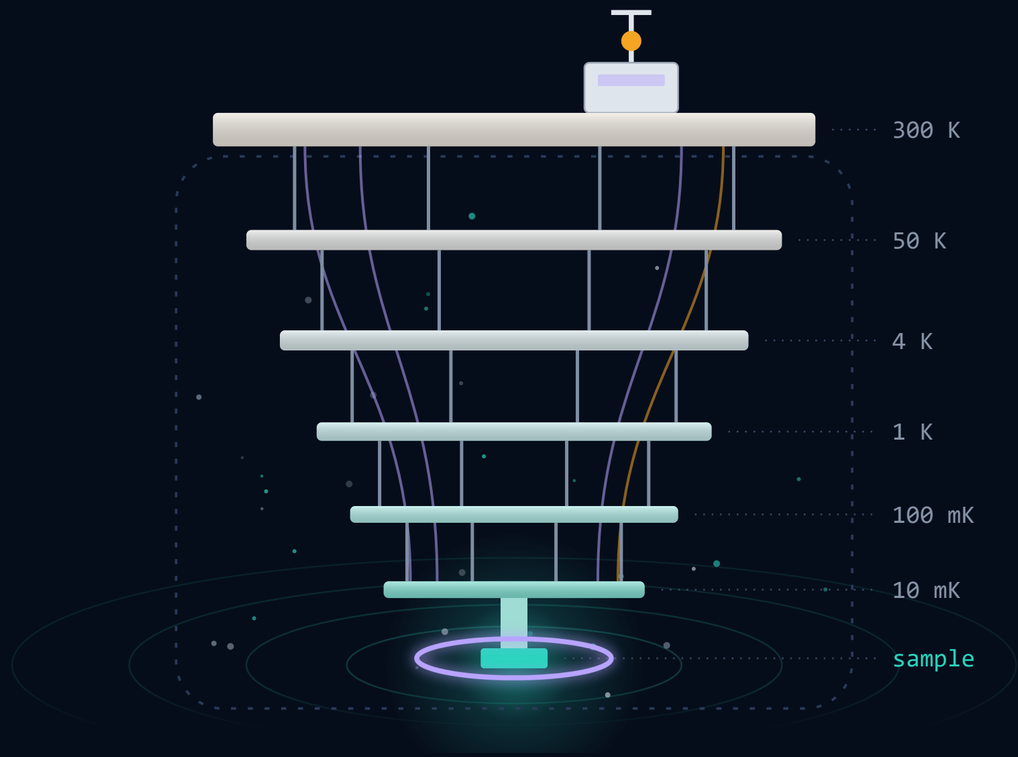
VIBECHECK · THE USABLE FORMAL METHODS
HACKATHON · SAN FRANCISCO · NOV 2026

CryoLean

A cryostat checker where every verdict is a theorem.

Lean 4 · decides **Z3 · Claude · Lanyon · propose**

Forall · audits



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hrahman12.github.io/CryoLean · github.com/hrahman12/CryoLean · generic cryostat drawn in code, no real model

/ THE THEOREMS · LEAN 4

If Lean says met, every requirement holds. Provably.

THE THEOREMS · VERBATIM FROM LEAN/CRYOLEAN/

```
theorem verdict_accept_iff … : verdict cfg e = .accept ↔ AcceptSpec cfg e
```
→ **met** exactly when the specification holds

```
theorem verdict_reject_sound … (h : verdict cfg e = .reject) : RejectSpec cfg e
```
→ every **does not meet** rests on a documented violation

```
theorem fail_needs_trusted … (h : checkCapability k c = .fail) : TrustedEvidence c.evidence
```
→ every failure rests on trusted evidence

```
theorem verdict_refines (hr : Refines cfg cfg') (hd : verdict cfg e ≠ .needMoreInfo) : verdict cfg' e = verdict cfg e
```
→ new data never flips a decided verdict

128
theorems · 0 sorry / admit / native_decide

48
axiom-audited theorems, standard axioms only

0
dependencies · exact rationals, no floats

lean/CryoLean/Soundness.lean · Monotone.lean · lean/audit-axioms.json · Lean 4.35.0, no Mathlib · rationals with 7 SI dimensions · binders elided at …

/ FORENSICS · AUDITING THE CHECKER

We audited our own verifier. It found a real bug.

CHECKCAPABILITY · CHECKER.LEAN:65-73 · “NOT INSTALLED” RECORD

BEFORE

```
if k = .flagIs false
  then (if trustedB c.evidence then .pass else .unknown)
  else .fail    -- even with no source
```

AFTER

```
if trustedB c.evidence
  then (if k = .flagIs false then .pass else .fail)
  else .unknown
```

The build was green: the spec had the same hole. Now proved:

`fail_needs_trusted` · `untrusted_never_passes` · `untrusted_never_fails`

839

declarations scanned: 0 non-standard axioms, 0 smells, 0 unsafe flags

27

flags from 7 judges, each re-checked by the Lean kernel

1

real soundness gap, fixed

docs/forensics/README.md · lean/CryoLean/Checker.lean:65-73 (fix b0b02d5) · the scan predates Monotone.lean · tool run by the team, not a third-party audit

/ UNTRUSTED HELPERS, ONE JUDGE

AI proposes. Lean decides.

UNTRUSTED · PROPOSE

Claude
matches setups

Lanyon AI
physics proofs

Z3
alternatives, fixes

TRUSTED · DECIDES

Lean 4 checker

Each candidate gets Lean's verdict.

`candidates_verdicts_are_lean`

`Soundness.lean:1056`

THE VERDICT CARD

- **Requirements met**
- **Need more information**
- **Does not meet requirements**

`+ reason · theorem names · axioms · source hash`

2 of 7

proofs admitted by the Gatekeeper. Rejected: sorry, native_decide, a custom axiom, a false statement, an AI proof with 54 compile errors.

15,795 / 15,795

designs where Z3 and Lean agree

lean/CryoLean/Soundness.lean:1056-1069 · cryolean/gatekeeper.py · paper/data/e9_gatekeeper.json · Lean proves conclusions from the modelled requirements and the supplied, sourced facts; it does not prove a datasheet is correct or that an experiment will succeed.